

CODING ASSIGNMENT

1. Take User input and process

- Take 5 integer input from the user.
- Remove all numbers less than 9.
- Calculate the sum of remaining numbers
- Commit this code on your github link under basic branch

2. Take User input and decode logic

- Take 2 integer input from user and print their products(their multiplication)
- IF product is greater than 500 then return their sum
- If the product is smaller than 500 then return "Hello SPSU code is running fine !!"
- commit this code on your github link under basic branch

3. Take User input in string form and perform operation

- Take input from user in string form only and calculate the length of string
- IF length is greater than 7 then display only those character which are present in even index number
- if length is less than or equals to 7 then display only those character which are present in odd index number
- file name must be SPSUstrindex in which you are writing code
- commit this code on your github link under basic branch

4. dynamic array operations and decoding

- two dynamic arrays are given below L1, L2.

- create a new dynamic array called L3 containing items in below given pattern
- From L1 it must take only odd index items
- From L2 it must take only even index items
- filename must be SPSUdynamic array in which you are going to write the code
- commit this code on your github link under basic branch

Inputs : L1 = [11, 21, 24, 12, 18]

L2 = [14, 44, 25, 37, 13]

Expected output : L3=[11,24,18,44,37]

5. Attendance of the Employees

- A Company's CEO wants the attendance of all the employees
- There are e no.of employees and w no.of working days
- Find the maximum number of consecutive days on which all the employees are present
- File name must be SPSUattendance in which you are writing the code
- Commit this code on your github link under basic branch

* Constraints:

$1 \leq e \leq 10$

$1 \leq w \leq 31$

Each data[i][j]={'P','A'}

Sample Input: e= 5 , w= 7

Data = [PPPPP, PPAPP, AAAPP, PAPAP, AAAAA, PAAAP, PPPPP]

Sample output: 1,7

6. Splitting the dynamic array

- Create a dynamic array ranging from 1-20
- Create a new dynamic array that contains first and the last five elements of the existing dynamic array
- Now create and display another dynamic array that contains square of the elements of the new dynamic array
- Now split the new dynamic array into three parts where the length of splitted parts should be two, three and five respectively
- File name must be in SPSUdynamicarray in which you are writing code
- Commit this code on your github link under the string branch

7. Raining like cats and dogs

- Take user input in the form of a string.
 - Return True if the string "cat" and "dog" appear the same number of times in the given string
 - file name must be SPSUcatndog in which you are writing code
 - commit this code on your github link under string branch
- Sample Input Sample Output

'Catdog' True

'Catcat' False

'Hello' False

'1cat1cadodog' True

8. Convert tuple to dynamic array using given string

- Take input from the user in the form of tuple
- Take another input from the user in the form of string
- Convert the tuple to dynamic array
- Add string to the dynamic array after each element
- Convert dynamic array to tuple again
- File name must be SPSUconvtuple in which you are writing the code
- Commit this code on your github link under basic branch

Sample Input: (21,37,18), K= "Age"

Sample output: (21,"Age",37,"Age",18,"Age")

9. Rearranging the dynamic arrays

- Take numeric input from the user in a form of dynamic array of tuples
- Rearrange the tuples according to the number of elements in ascending order of tuples to obtain the new dynamic array
- If the dynamic array consists of tuples with equal elements then arrange them according to their sum of elements
- If sum of their elements is equal then arrange them according to their first least number

- Display the new dynamic array
- File name must be in SPSUdynamic array in which you are writing code
- Commit this code on your github link under the python dynamic array branch

Sample Input: [(2,1,2), (2,5), (4,5,3,7), (3,1), (1,6)]

Sample Output: [(1,6), (2,5), (3,1), (2,1,2), (4,5,3,7)]

10. Paragraph to dynamic array

- Take input from the user in string form(a sentence or para)
- All the words in the string having more than 4 letters should be stored in a dynamic array
- file name must be SPSUstring_dynamic array in which you are writing code.
- commit this code on your github link under basic branch

Sample Input : A paragraph is defined as “a group of sentences or a single sentence that forms a unit”. Length and appearance do not determine whether a section in a paper is a paragraph.

Sample Output : ['paragraph' , 'defined' , 'group' , 'sentences' , 'forms' , 'Length' , 'appearance' , 'determine' , 'whether' , 'section' , 'paper' , 'paragraph']

11. Make a program to merge two dynamic array into a single dictionary/map

- Take inputs from the user
- Any one dynamic array must contain unique elements
- both the dynamic array should be of the same size
- both the dynamic array should be a combination of numbers and names
- Name of dictionary/map you can take it accordingly
- file name should be in SPSUmerge
- commit the code on github link under basic branch